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REVIEW ARTICLE

Snakebite envenoming in India: it is time we look beyond the concept of the Big Four species

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India faces substantial challenges from snakebite envenoming secondary to the high morbidity, mortality and financial burden, particularly in rural communities. While concentrated on the 'Big Four' venomous species, recent research indicates a necessity to expand the focus to encompass additional medically relevant species. This review emphasizes the geographic heterogeneity in venom among these snakes, which impacts antivenom effectiveness and necessitating region-specific formulations. This analysis highlights the shortcomings of current antivenoms and identifies non–Big Four species involved in snakebite envenoming, advocating for an urgent shift to inclusive antivenom strategies that integrate local venom profiles to enhance treatment effectiveness and thereby reduce snakebite-related morbidity and mortality. Improved training for healthcare providers and enhancements in anti-snake venom quality are essential for meeting the World Health Organization's 2030 Sustainable Development Goal objective of halving snakebite-related fatalities and disabilities. Incorporating snakebite management into national health programs and conducting epidemiological research systematically are crucial to mitigating this preventable health concern.

Keywords: disability, ecosystems, morbidity, mortality, snakebite envenoming, WHO SDG

Introduction

India accounts for approximately half of the global snakebite deaths, with an estimated 58 000 fatalities annually.[1] The country is home to >310 snake species, of which about 60 are venomous.[2] The majority of snakebite incidents occur in rural communities, particularly among agricultural workers and other outdoor occupations.[3]

Traditionally, the 'Big Four' snakes—the Indian cobra (*Naja naja*), common krait (*Bungarus caeruleus*), Russell's viper (*Daboia russelii*) and saw-scaled viper (*Echis carinatus*)—have been considered as the species responsible for the majority of snakebite envenoming in India. However, recent studies suggest that this classification may no longer fully encompass the current snakebite scenario in India, especially in the context of certain geographies. This article examines the relevance of the Big Four concept, the significance of non–Big Four species, geographical venom variation and antivenom efficacy and proposes recommendations for improved snakebite management.

Is the Big Four concept still relevant?

The Big Four concept was created to standardise the production of polyvalent antivenom in India, focussing on four species thought to account for the majority of venomous bites. Although these snakes are major contributors to snakebite occurrences, recent research suggests that other species are also relevant as a cause for morbidity and mortality. For example, the hump-nosed pit viper (*Hypnale hypnale*) is acknowledged as a major offending snake in Kerala and parts of Karnataka.[4]

Several non–Big Four species have been implicated in medically significant envenoming. These species contribute notably to the snakebite burden (Figure 1), yet their venoms are poorly studied and no specific antivenom is available for treatment.[4]

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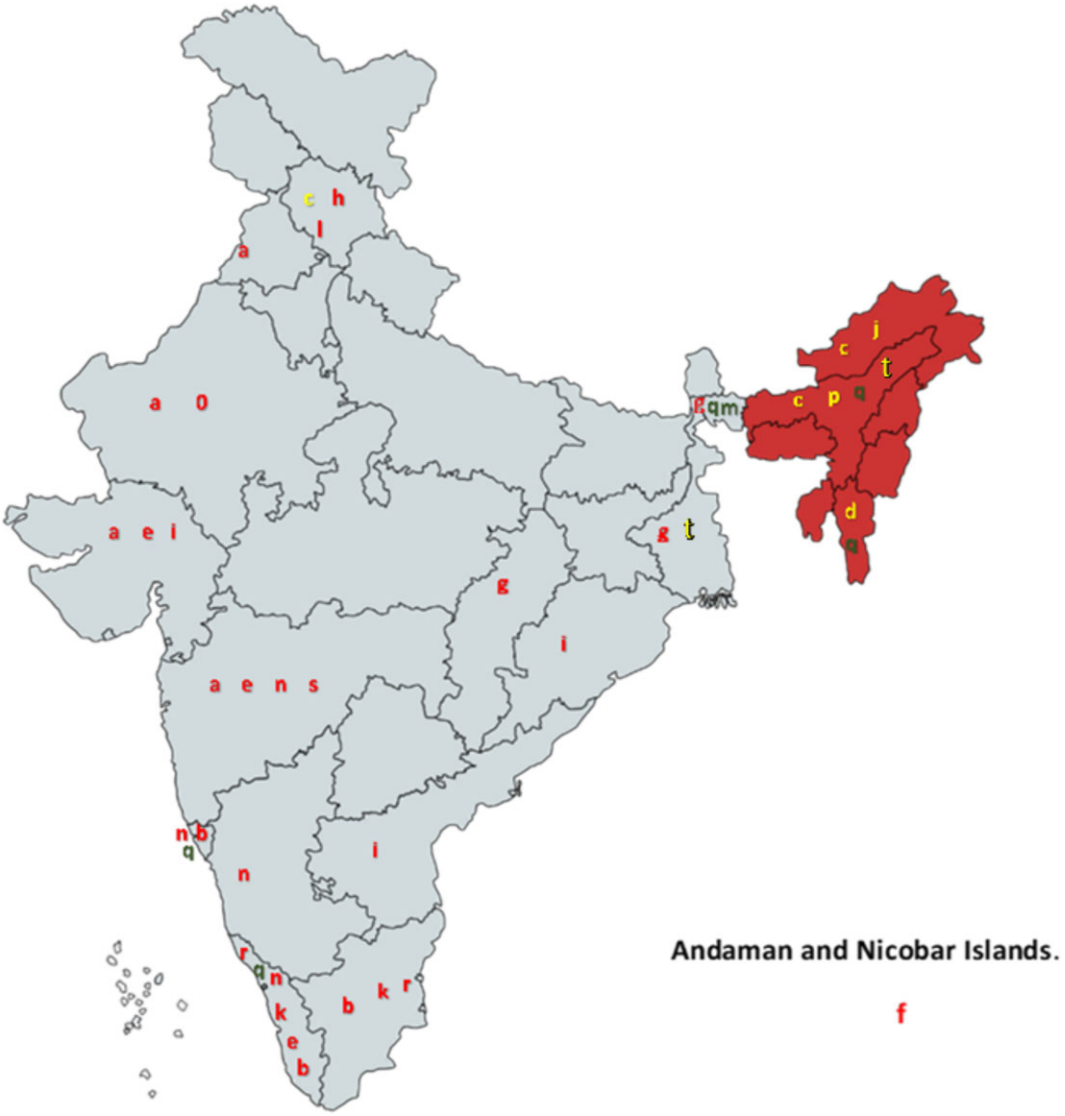

Figure 1. Distribution of non–Big Four species across Indian states.[5] This map illustrates the distribution of key non-Big Four snake species identified by serial alphabets as listed in Table 1. *The red-coloured states indicate where non–Big Four snake species contribute more significantly to envenomation incidents than the Big Four species.[6,7]

The hump-nosed pit viper, *H. hypnale*, is a medically significant yet frequently overlooked species in India. Historically, *H. hypnale* envenomation was thought to induce only localized effects, however, contemporary research confirms its ability to cause systemic toxicity, including venom-induced consumption coagulopathy, acute kidney injury and thrombotic microangiopathy.[8,9] The Indian polyvalent antivenom fails to neutralize *H. hypnale* venom, resulting in ineffective treatment and unnecessary antivenom administration[10,11]

A retrospective study conducted at a tertiary hospital in Kerala indicated that among the 47.3% of cases with known snake species, *D. russelii* accounted for 23.1%, *H. hypnale* accounted for 15.4%, *N. naja* accounted for 5.5% and *B. caeruleus* accounted for 3.3%.[12] Additionally, data from online snake rescue databases such as indiansnakes.org corroborate these findings, highlighting the frequent encounters with *H. hypnale* in these regions. Significantly, hospitals in Kerala, Tamil Nadu and Karnataka refrain from administering polyvalent antivenom when *H. hypnale* is accurately identified as the causative agent.[8,12,13]

The Sind krait (*Bungarus sindanus*), which has an extensive distribution across Pakistan, Rajasthan, Gujarat and Maharashtra, is responsible for both fatalities and long-term morbidity. Due to morphological similarities with *B. caeruleus*, a significant number of *B. sindanus* envenomings are misdiagnosed as common krait bites, leading to suboptimal treatment.[14,15] A commercial polyvalent antivenom from one Indian manufacturer demonstrated moderate cross-reactivity with *B. sindanus* venom sourced from Sindh, Pakistan, though neutralization was incomplete.[14,15]

E. carinatus sochureki, referred to as Sochurek's viper, is the primary agent of envenomation in the Thar Desert, which extends

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Table 1. Non–Big Four venomous snakes of India.[2]

Sl ID.	Species (scientific name)	Key identification features	Geographic distribution	Neutralization by current polyvalent ASV
a	Sind krait (*Bungarus sindanus*)	Medium-sized krait; glossy black body with narrow white bands; head barely distinct from neck	Northwestern India: Rajasthan, Punjab, Gujarat	Poorly neutralized
b	Hump-nosed pit viper (*Hypnale hypnale*)	Small, stout; pointed, upturned snout forming a 'hump'; tail tip often yellow or reddish	Southwestern India: Western Ghats, Kerala, Karnataka and Konkan	Not neutralized
c	Salazar's pit viper (*Trimeresurus salazar*)	Small arboreal pit viper; bright green body, males have a reddish-orange stripe	Northeastern India: Arunachal Pradesh	Not neutralized
d	Mountain pit viper (*Ovophis monticola*)	Stout terrestrial viper with a short snout; grey-brown with darker brown blotches	Himalayas, northeast India	Not neutralized
e	Hook-nosed sea snake (*Enhydrina schistosa*)	Aquatic; distinctive 'beaked' snout; adult coloration usually uniform dark grey above	Indian Ocean coasts, Bay of Bengal, Arabian Sea	Not neutralized
f	Andaman's cobra (*Naja sagittifera*)	Medium-large spitting cobra; colour highly variable, often glossy black	Andaman Islands	Not neutralized
g	Banded krait (*Bungarus fasciatus*)	Large krait; alternating bands of yellow and black encircling the body	Eastern and northeastern India	Partially neutralized
h	Himalayan pit viper (*Gloydius himalayanus*)	Small pit viper; light brown or grey with darker brown blotches	Western Himalayas	Not neutralized
i	Annulated sea snake (*Hydrophis cyanocinctus*)	Large sea snake; pattern of annular black rings on a bluish background	Bay of Bengal, Andaman Sea	Not neutralized
j	Greater black krait (*Bungarus niger*)	Medium-sized krait; predominantly jet black	Eastern Himalayas, northeast India	Not neutralized
k	Large-scaled pit viper (*Craspedocephalus macrolepis*)	Bright green arboreal pit viper; large, overlapping dorsal scales	Southern Western Ghats, India	Not neutralized
l	Central Asian cobra (*Naja oxiana*)	Medium-large cobra; uniform light brown, no hood mark	Northwest India: Jammu and Kashmir	Not neutralized
m	Lesser black krait (*Bungarus lividus*)	Small, slender krait; almost solid black	Northeastern India	Not neutralized
n	Malabar pit viper (*Craspedocephalus travancoricus*)	Variable coloration; stout body with broad triangular head	Western Ghats, India	Not neutralized
o	Sochurek's saw-scaled viper (*Echis carinatus sochureki*)	Similar to common saw-scaled viper; larger and paler	Northwestern India	partially neutralized
p	Red-necked keelback (*Rhabdophis subminiatus*)	Small; bright red/orange patch on the neck	Northeast India	Not neutralized
q	King cobra (*Ophiophagus hannah*)	World's longest venomous snake; olive-green, brown, or black body	Southwest and northeast India	Negligible neutralization
r	Northern large-scaled pit viper (*Craspedocephalus macrolepis*)	Identical to large-scaled pit viper; green colour with lateral yellow/white stripe	Southern Western Ghats, India	Not neutralized
s	Short sea snake (*Hydrophis curtus*)	Short, thick-bodied sea snake; bold black bands around a lighter body	Indian coastal waters	Not neutralized
t	Indian monocled cobra (*Naja kaouthia*)	About 1.5 m in length, usually light brown or yellow with a distinct 'monocle' pattern that is a circular or oval ring on the hood. Spreads a wide hood when threatened, showing the distinctive monocle pattern	Northeast India, especially in Odisha Assam, West Bengal and parts of the northern plains	Poorly neutralized

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through Rajasthan and portions of Gujarat. Nonetheless, the existing Indian polyvalent antivenoms demonstrate inadequate cross-reactivity with *E. carinatus sochureki* venom, hence generating concern regarding treatment efficacy in this area.[16] A study in western Rajasthan revealed that the clinical efficacy of the present Indian polyvalent antivenom is insufficient for treating envenomation caused by *E. carinatus sochureki*. The study revealed a notable incidence of clinical bleeding and delayed coagulopathy in patients, with a large percentage unresponsive to the provided antivenom.[17]

The banded krait (*Bungarus fasciatus*), recognised for its very toxic venom, infrequently results in human envenomation owing to its elusive behaviour. Fatal events are few, occurring in regions such as Odisha and West Bengal. Certain antivenoms have only limited efficacy, and their trustworthiness remains unvalidated in comprehensive clinical trials.[18] Venom analysis revealed substantial geographical disparities in venom, influencing antivenom effectiveness and indicating the necessity for region-specific antivenoms to enhance treatment outcomes.[19]

The greater black krait (*Bungarus niger*), a species found in Uttarakhand, Assam, Sikkim and Arunachal Pradesh, has been associated with multiple envenomation incidents and fatalities in these regions. Nevertheless, similar to other non–Big Four species, its venom remains unneutralised by current polyvalent antivenoms, resulting in unsatisfactory treatment outcomes.[20]

Sea snake envenomations, though less prevalent than terrestrial snakebites, frequently occur when fisherman interact with snakes trapped in fishing nets. A significant number of incidents remain unreported, particularly in isolated coastal regions. Although infrequent, sea snake bites can result in significant morbidity and death owing to their strong neurotoxins.[21] The hook-nosed sea snake (*Enhydrina schistosa*) is responsible for the majority of documented bites, however, existing polyvalent antivenoms inadequately neutralise its venom.[22]

A hospital-based study conducted at the Demow Community Health Centre in Assam documented 1011 snakebite cases, of which 139 (13.7%) were identified as venomous bites. Among these, viperid snakes (green pit viper and Salazar's pit viper) were responsible for 66.19% of cases, while elapid species (Indian monocled cobra [*Naja kaouthia*], *B. fasciatus* and *B. niger*/lesser black krait [*Bungarus lividus*]) accounted for 21.5%. None of the venom from these species is effectively neutralised by the currently available polyvalent anti-snake venoms (ASVs).[23]

These findings highlight the critical necessity for region-specific antivenom formulations capable of effectively neutralising the venom of snake species native to various biogeographical zones in India. Furthermore, contemporary antivenom production methods must evolve to include a wider array of venom samples sourced from various locations, thereby improving the clinical efficacy of snakebite therapies across the nation.

Biogeographical venom variation and its impact on treatment

Venomous snakes are taxonomically categorized into four distinct families: Elapidae, Colubridae, Viperidae and Atractaspididae. The Viperidae family includes vipers and pit vipers, while species such as cobras and kraits belong to the Elapidae family, which also encompasses sea snakes classified under the subfamily Hydrophiinae. The Colubridae family contains rear-fanged snakes (opisthoglyphous species). The Atractaspididae family is less common in India.[24]

The pathological effects of snake envenomation are varied, including neuromuscular paralysis (neurotoxicity), systemic haemorrhage and coagulation abnormalities (haemotoxicity), localised oedema, blistering and tissue necrosis (cytotoxicity) at and around the bite site.[25] The geographical variability in venom toxicity and neutralisation with commercially available ASVs results from significant heterogeneity in venom composition, occurring both between species (interspecific) and within the same species (intraspecific) and is well documented.[4,26–29] The biochemical heterogeneity in venom needs to be accounted for before assessing the efficacy of snakebite treatment with antivenoms. The requirement therefore is for the manufacture of ASV specific to the geography considering the heterogeneity of venom between regions.[30]

The composition of venom is thought to be affected by ecological and environmental factors, including nutritional adaptations, predation pressures, climate and ontogenetic changes across various geographical regions.[31–35] Significantly, similar variances have been noted not only across vast geographic expanses, but also across rather short distances.[4,36] Venom diversity is a crucial factor in ecological specialisation among snakes, however, it presents considerable hurdles in clinical care, especially for antivenom effectiveness. Indian polyvalent antivenom is produced by hyperimmunising equines with venom extracted from the Big Four species, followed by purification of the resultant antibodies. However, the venom samples used in this process are predominantly sourced from a limited number of districts in southern India, raising concerns about their efficacy against snake populations from other biogeographic zones, where venom composition may differ considerably.[18,37]

Venom composition varies not only between species, but also regionally within the same species, a phenomenon known as venom variation. This has significant implications for antivenom efficacy. For example, studies have shown that Russell's viper venom in northern India differs significantly from that in southern India in terms of toxicity and enzymatic activity. Similarly, the venom of monocled cobras varies geographically, affecting the clinical presentation of envenomation and response to antivenom.[38]

A study employing various venom profiling techniques including sodium dodecyl sulphate polyacrylamide gel electrophoresis, electrospray ionization liquid chromatography tandem mass spectrometry and proteomic analyses demonstrated notable differences in venom composition among *N. naja*, *B. caeruleus* and *E. carinatus*. Several distinct protein bands of varying molecular weights and intensities were observed across these species, highlighting significant biochemical variations not only between species, but also among different populations of the same species.[4]

Compositional and biochemical analyses of venoms in a pan-Indian study on *N. naja* revealed substantial pathological variability in envenomation cases across different regions. The pharmacological characteristics and efficacy of venoms derived from geographically diverse populations of *N. naja* exhibited considerable diversity, raising substantial concerns for therapeutic management. The research additionally

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verified that commercially available antivenoms demonstrate restricted neutralisation effectiveness against these local venom variations.[39]

Similarly, investigations into *D. russelii* venom collected from five distinct biogeographic zones in India revealed considerable intraspecific variation in its composition, biochemical activity and pharmacological potency. These differences directly influence clinical outcomes and the effectiveness of available antivenoms.[40]

Venom of *E. carinatus* sourced from three geographically distinct populations exhibited unique reversed-phase high-performance liquid chromatography profiles, with toxin peaks eluting at comparable time intervals but displaying differences in peak areas. This suggests notable variations in the relative concentration of specific toxins, which could impact the venom's pathophysiological effects and the effectiveness of antivenom therapies.[41]

Venom variation and its influence on clinical efficacy was used to assess the performance of antivenoms against *B. caeruleus*, *B. sindanus* and *Bungarus walli* (Wall's krait). While manufacturers claim effective neutralization against *B. caeruleus*, findings revealed poor efficacy against the venoms of *B. sindanus* and *B. walli*.[42] In vivo venom neutralization assays further demonstrated that even the best-performing commercial antivenoms failed to achieve the marketed potency levels against geographically distinct krait populations.[25]

Efficacy of current antivenom in India

Indian polyvalent antivenom is produced by utilising venom from the Big Four snakes and obtained from a restricted geographic area. Nonetheless, venom composition exhibits considerable variation among distinct populations, even within the same species, resulting in diminished efficacy when addressing bites from snakes outside these areas. Consequently, numerous snakebite incidents do not adequately respond to antivenom, leading to treatment failures, extended hospitalisations and heightened morbidity and fatality rates.[43] This issue is further compounded by the lack of coverage for non–Big Four species, such as *H. hypnale* and *N. kaouthia*), whose venom remains largely unaffected by existing formulations. Consequently, physicians often resort to administering excessive doses of antivenom, increasing the risk of severe adverse reactions and antivenom-associated complications. A systematic review has further emphasized the limited efficacy of polyvalent antivenom beyond the Big Four species, underscoring the urgent need for region-specific antivenoms to enhance treatment outcomes.[43]

The World Health Organization underscores the critical importance of developing antivenoms that are specifically tailored to the venom profiles of medically significant snake species, particularly those prevalent in distinct geographic regions. This approach involves a comprehensive characterization of the venom proteome of these species, followed by rigorous preclinical testing to assess the neutralisation efficacy of the antivenoms. Such region-specific antivenoms are essential to enhance treatment effectiveness and reduce the mortality and morbidity associated with snakebites.[44]

Challenges and government initiatives in establishing regional venom collection centres in India

The establishment of regional ASVs in India is hindered by various constraints, such as prohibitions on handling certain venomous species, restricted permits for venom collection from a limited number of snakes and delays in acquiring essential approvals and sampling permits. These regulatory obstacles impede the collection of representative venom samples from various snake populations, thus impacting the formulation of region-specific antivenoms.

Establishing regional venom collecting centres is essential for the development of region-specific antivenoms; nevertheless, many hurdles impede advancement. The geographical diversity complicates the acquisition of representative venom samples, necessitating substantial fieldwork. The restricted facilities, characterised by a singular prominent venom collection centre in Tamil Nadu, engender logistical challenges in the collection and distribution of venom.[45]

Regulatory hurdles slow expansion, as approvals involve wildlife conservation laws and ethical considerations.[2] Financial constraints further impede progress, as venom collection requires substantial investment, with snakebite envenomation predominantly affecting rural, economically disadvantaged populations.[46] Additionally, a shortage of trained herpetologists and venom handlers limits operational efficiency.[2]

To address these challenges, the National Action Plan for Prevention and Control of Snakebite Envenoming was launched in March 2024, aiming to reduce snakebite deaths by 50% by 2030 by enhancing venom collection and antivenom production.[2] The government plans to establish regional venom centres, ensuring the availability of locally effective antivenoms.[47] Legislative support is also being strengthened to streamline regulatory approvals and simplify expansion efforts.[2] The government of India has advised all states to classify snakebite cases and fatalities as a 'notifiable disease' under the appropriate regulations of State Public Health.[48]

Proposed solutions and the way forward

Improving antivenom production

Developing region-specific antivenoms is crucial to accommodate venom variances among diverse snake populations. Enhancing research on non–Big Four venomous species will facilitate the development of broad-spectrum antivenoms, guaranteeing effective treatment for medically significant neglected species. The integration of sophisticated biotechnologies, including recombinant technology and monoclonal antibodies, can markedly improve the efficacy and safety of antivenom.

Strengthening snakebite management

Improving medical training is essential for advancing snakebite diagnosis and treatment, particularly for non–Big Four species that are frequently misidentified. Establishing regional poison control centres with specialised knowledge of snakebite treatments will enable prompt and efficient treatment. Furthermore, the creation of species-specific point-of-care diagnostics would

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enhance the precision of envenomation case identification, facilitating focused and prompt intervention. Incorporating snakebite management into national health programs and performing localised epidemiological research are crucial for alleviating this preventable health concern.

Public awareness and prevention

Conducting educational initiatives in remote and high-risk regions will enhance understanding regarding snakebite avoidance, appropriate first aid techniques and the necessity of prompt medical attention. Enhancing community awareness and empowerment and supplying protective equipment, including appropriate footwear and bed nets, can markedly decrease the incidence of snakebites in at-risk groups.

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